

Paper 1

Title

Dynamic and Explainable Fish Mortality Prediction under Low-Concentration Ammonia Nitrogen Stress

Abstract

Fish mortality caused by deteriorating water quality is one of the major challenges in intensive aquaculture. Among various environmental factors, ammonia nitrogen is highly toxic and significantly affects fish survival even at relatively low concentrations. Traditional statistical methods are unable to accurately capture the dynamic relationship between environmental conditions and fish mortality. This study proposes a dynamic fish mortality prediction model using Long Short-Term Memory (LSTM) networks that combine static fish characteristics with cumulative time-series environmental data. Water quality parameters and operational information were collected continuously from aquaculture ponds and used to train the deep learning model. To improve the transparency of the prediction process, SHAP (Shapley Additive Explanations) was employed to identify the contribution of each feature over time. Experimental results demonstrated that the proposed model achieved high prediction accuracy, with an external validation coefficient of determination (R^2) of approximately 0.81 while effectively adapting to changing aquaculture conditions. The study concludes that integrating dynamic time-series learning with explainable AI provides a reliable decision-support tool for predicting fish mortality under ammonia stress and assists farmers in taking timely preventive actions.

Introduction

Aquaculture has become an important contributor to global food production, making fish health management increasingly essential. Among the environmental factors affecting fish survival, ammonia nitrogen is one of the most hazardous pollutants because prolonged exposure, even at low concentrations, causes stress, weakens fish immunity, and increases mortality. Conventional mortality prediction methods generally rely on static measurements collected at a single point in time and therefore cannot effectively represent continuously changing aquaculture environments.

Recent developments in machine learning have enabled the analysis of complex environmental data collected from modern aquaculture monitoring systems. Long Short-Term Memory (LSTM) networks are particularly suitable for modeling sequential environmental data because they can capture long-term temporal relationships among water quality parameters. In this study, the authors performed ammonia toxicity experiments to determine lethal concentration values and collected static fish characteristics together with dynamic environmental measurements such as dissolved oxygen, temperature, pH, turbidity, ammonia concentration, and operational information. These

data were combined to train an LSTM-based prediction model capable of estimating hourly fish mortality under low-concentration ammonia stress.

To overcome the "black-box" nature of deep learning, the study incorporated SHAP analysis to explain how different environmental variables influenced mortality predictions over time. The proposed framework demonstrates that combining dynamic prediction with explainable AI can significantly improve decision-making in intensive aquaculture systems.